ICT2142 · Database Systems

Worked Examples: SQL Joins and Aggregation

Original synthetic reading for demonstration and independent study. It is not an official syllabus, lecture note, or assessment specification.

Learning objectives

After this reading, a learner should be able to explain the central idea, apply it to a small example, and identify at least one assumption or limitation.

1. Main idea

Joins combine related rows using matching keys. Grouping computes summaries over sets, and HAVING filters groups after aggregation. NULL represents missing or unknown information and requires IS NULL rather than equality comparison.

2. Example and reasoning

To list enrolled students with course codes, join enrollments to students and offerings, then offerings to courses. A LEFT JOIN is useful when the report must retain courses that currently have no enrollments.

3. A method for practice

1. State the problem and define the terms that affect the answer.
2. Write down the assumptions and information available.
3. Work through a small example and make each step visible.
4. Check the result against a second case or independent explanation.
5. Summarize what the result supports and what it does not establish.

4. Questions for study

- Explain “worked examples: sql joins and aggregation” in your own words and give a new example.
- Which assumption matters most in the example, and how would the answer change if it were false?
- Describe one common mistake and a practical way to detect it.
- Write a short checklist another learner could use to review this work.

Reflection

Separate evidence from assumptions. If a conclusion depends on information not provided here, identify what would need to be verified.